

lys_instr: A Python Package for Automating Scientific Measurements

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Summary

Modern experiments increasingly demand automation frameworks that coordinate diverse scientific instruments while remaining flexible and customizable. Developing custom measurement systems, however, often requires explicit management of low-level device communication and concurrency, resulting in substantial development overhead. We present `lys_instr`, a Python package that addresses these challenges through an object-oriented, three-layer architecture for scan-based instrument control, workflow coordination, and GUI construction. It enables researchers to construct responsive, asynchronous measurement systems by reusing instrument abstractions, workflow logic, and GUI components across hardware configurations. Integrated with the `lys` platform (Nakamura, 2023), `lys_instr` provides a common environment for experiment control, data acquisition, and multidimensional visualization.

Statement of need

Many scientific experiments require measurements across wide parameter spaces to elucidate physical phenomena. As experiments involve longer acquisition times and more diverse instruments, efficient automation becomes increasingly valuable. Measurement workflows may also incorporate informatics-driven, condition-based optimization through external Python libraries, motivating reusable programmatic interfaces. `lys_instr` focuses specifically on software-coordinated control-and-detect workflows.

Building such systems repeatedly incurs costs at both the device and application levels. Hardware-specific communication must be combined with state monitoring and asynchronous execution, while scan logic, storage, visualization, and operator controls are often rebuilt for individual experiments. `lys_instr` addresses this overhead through common controller and multidimensional detector interfaces, together with reusable workflow and GUI components defined against those interfaces. GUIs complement programmatic control by providing real-time visualization, device-state monitoring, and manual intervention when human judgment is required during acquisition. The Python-based interfaces allow instrument control to be combined with `lys` and other Python libraries for scientific computing and informatics. Reusing these shared components allows users to concentrate their implementation effort on communication with specific instruments and procedures unique to each experiment.

State of the field

Scientific instrument-control software spans environments with different priorities. LabVIEW (National Instruments Corporation, 2024) and MATLAB's Instrument Control Toolbox (The MathWorks, Inc., 2024) integrate instrument communication, workflows, and GUI development. QCoDeS (Nielsen et al., 2025) and PyMeasure (PyMeasure Developers, 2024) provide instrument abstractions and experimental procedures. PyMoDAQ (Weber, 2021) combines actuator and detector plugins with dashboards and scan extensions, while Bluesky (Allan et al., 2019) separates hardware abstraction from reusable orchestration plans and optional GUI clients. Qudi (Binder et al., 2017) explicitly separates hardware, experiment logic, and GUI layers. Atomize (Melnikov et al., 2025) and fsc2 (Törring, 2019) support script-based experiments through reusable device operations. Together, these systems establish hardware, workflow, and GUI abstractions but differ in how they integrate them and which workflows they target.

lys_instr focuses on software-coordinated control-and-detect workflows. Its three-layer architecture organizes controller and detector abstractions, reusable workflow and GUI logic, and system assembly under a consistent API. Developing lys_instr as a distinct package allowed this API to span all three layers while supporting direct integration with the Python-based lys platform. This integration combines instrument control with live multidimensional visualization and analysis, including MultiCut, in the same environment. Users add hardware-specific communication and specialized procedures through custom components.

Software design

The three layers and their relationships are summarized in Figure 1. Each layer applies established object-oriented design patterns (Gamma et al., 1994) to define how components are extended, composed, and connected.

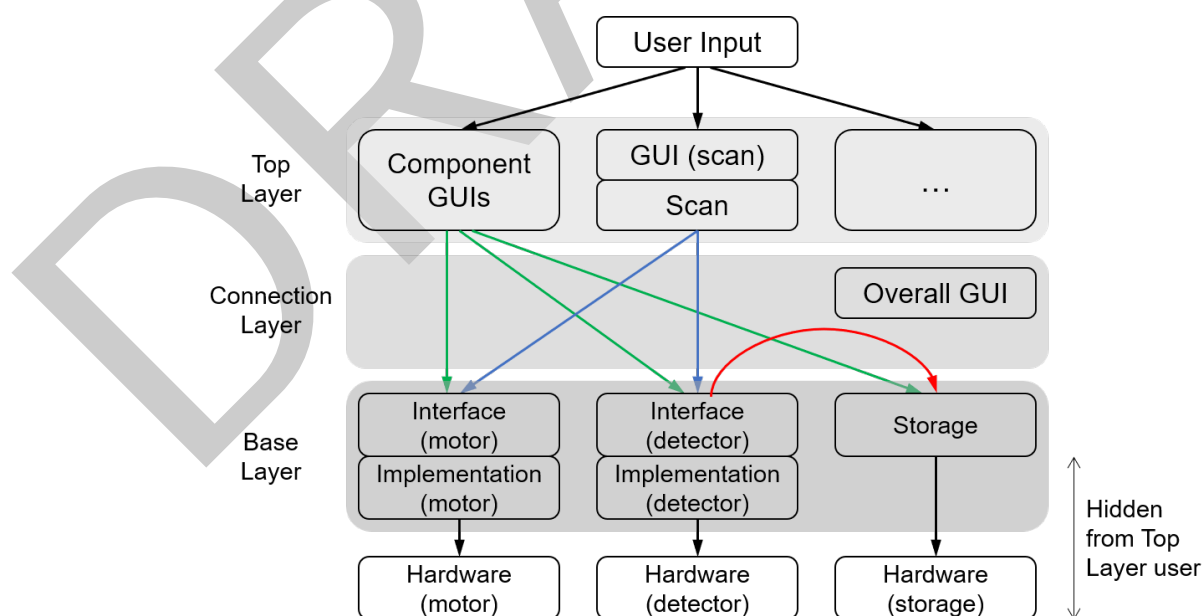


Figure 1: Schematic of the code architecture of lys_instr.

1. Base Layer: Instrument Abstraction

This layer defines standardized abstract interfaces for two principal roles in a control-and-detect workflow: *controllers*, which adjust experimental parameters such as external fields,

temperature, or physical position, and *detectors*, which acquire data such as images or spectra. Following the *Template Method* pattern, the base interfaces define the device lifecycle, state monitoring, synchronization, and asynchronous execution, while subclasses implement the primitive operations required for device-specific communication. Higher layers interact with different instruments through the same methods, independently of their concrete implementations. Because background monitoring and acquisition are managed by the interfaces, other devices and GUIs can remain responsive, and device-specific subclasses can be integrated directly into higher-level workflows without reimplementing concurrency logic.

2. Top Layer: Workflow Coordination

This layer constructs hardware-independent workflows from the Base Layer interfaces. In a *scan*, controllers vary experimental parameters while detectors acquire data at each step. Following the *Bridge* pattern, scan execution depends on controller and detector abstractions rather than concrete hardware implementations. The *Composite* pattern allows individual scan processes to be nested into multi-parameter workflows while retaining a common execution model. Prebuilt GUI components invoke the same abstract methods for direct device control and receive state and data updates through event-driven signals, following the *Observer* pattern and avoiding direct dependencies on concrete devices. Consequently, workflow and GUI logic can be reused across different hardware configurations without rebuilding standard control and coordination components.

3. Connection Layer: Control-System Assembly

This layer assembles components from the Base and Top Layers into an interactive control system, including device interfaces, workflows, data storage, visualization, and GUI components. Following the *Mediator* pattern, it centrally manages connections within and across layers, linking GUI components to their corresponding interfaces and coordinating data flow without direct dependencies among individual components. Concrete hardware details therefore remain confined to the Base Layer, allowing system configurations and GUI layouts to be adapted without application-level implementation of inter-device communication or thread coordination. Prebuilt templates provide starting configurations that can be adapted to individual experimental setups.

Together, these layers allow new instruments to be integrated through device-specific subclasses while reusing asynchronous execution, workflow coordination, and GUI behavior. Workflows with different coordination models can be supported through additional components or customized interfaces.

Example GUI configuration

`lys_instr` allows users to assemble GUIs suited to individual experimental setups; Figure 2 shows one possible configuration. Here, the `lys_instr` window is embedded in the `lys` platform, with Sector A for data storage, Sector B for detector control, and Sector C for controller operation and scan configuration. Multidimensional, nested scan sequences can be defined through the Scan tab, while `lys` tools in the outer window tabs allow on-the-fly customization of acquired-data visualization.

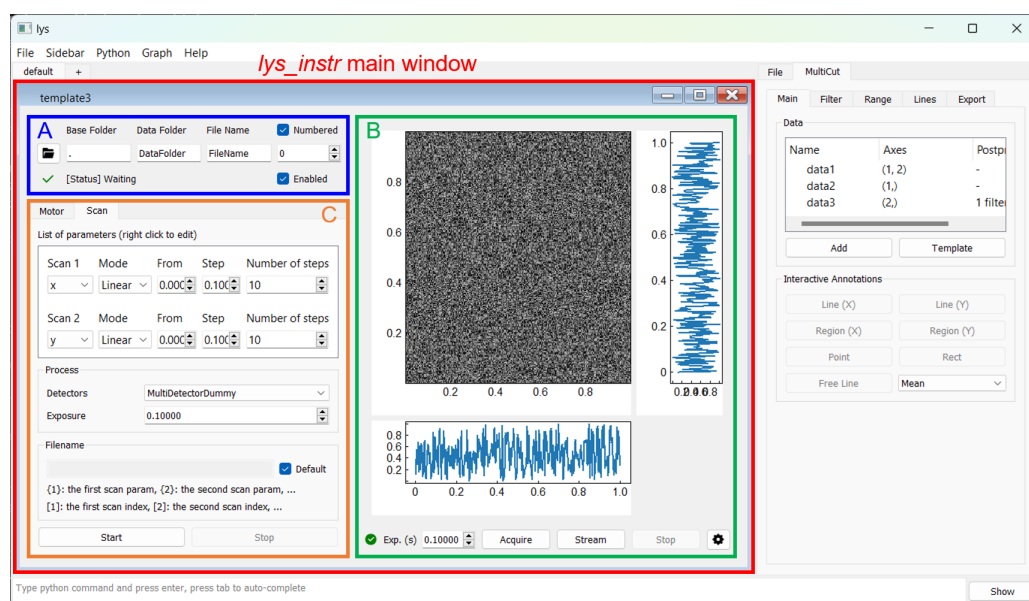


Figure 2: Example GUI configuration in `lys_instr`. The main window, embedded in the `lys` window, contains three sectors: storage (A), detector control (B), and controller operation and scan configuration (C). The Scan tab in (C) enables configuration of multidimensional, nested experimental workflows.

Research impact statement

`lys_instr` has been used in experiments reported in several peer-reviewed publications. It has been used to automate ultrafast electron diffraction (UED) and ultrafast transmission electron microscopy (UTEM) systems by coordinating experimental parameters and data acquisition in pump-probe experiments using ultrafast laser excitation and pulsed electron beams (Koga et al., 2024; Nakamura et al., 2020, 2021, 2022, 2023; Shimojima et al., 2021, 2023a, 2023b). It has also been used to control electromagnetic lenses and electron deflectors as part of complex microscopy workflows involving electron-beam precession (Hayashi et al., 2026; Shiratori et al., 2024).

The same interface-based workflows have been used to control transmission electron microscopes from multiple manufacturers at RIKEN Center for Emergent Matter Science and Nagoya University, demonstrating portability across hardware configurations. Integration with sister packages in the `lys` family, including `lys_em` (lys_em Developers, 2026) and `lys_fem` (lys_fem Developers, 2026), allows `lys_instr` components to participate in multi-instrument workflows while preserving modularity and extensibility.

AI usage disclosure

Generative AI tools provided debugging suggestions during the final stages of software development. The authors implemented and reviewed all changes. Core non-GUI behavior was tested with unit tests, while GUI and hardware workflows were validated with real instruments.

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